Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Operating System

BCA — Semester 3 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Why Optimal Page Replacement is best but not practically feasible page replacement algorithm?Calculate the number of page faults for Optimal, LRU and FIFO replacement algorithm for the reference string: 1, 3, 4, 2, 3, 5, 4, 3, 1, 2, 4, 6, 3, 2, 1, 4, 2 using 3 page frames.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Find Average waiting time and turn-around time for following example using FIFO, SRTF and Round Robin scheduling algorithm. Assume quantum as 4 ms.Process idArrival timeBurst time(ms)P108P215P3110P4213P5217

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Why virtual memory technique is used in the computer system? What is logical address? Explain the process of conversion of logical address to physical address using TLB.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Suppose that a disk drive has the cylinder numbered, 0 to 199 is currently serving a request at cylinder 143. The request queue is kept in the FIFO order 25, 17, 119, 197, 194, 15, 182, 115, and 183. What is the total head movement needed to satisfy these request for the following disk scheduling algorithm.a) FCFSb) SSTF

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. If a 2gb disk has 4-KB block size, calculate the size of the file allocation table if each entry of the table is 4 bytes.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is process? Explain 3 state process models with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Write down the basic difference between coalescing and compaction with diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is memory mapped I/O? Explain about device independent I/O software.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Differentiate between deadlock and starvation? Discuss the process of detecting deadlocks when there are multiple resources of each type.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Differentiate between process and a program. What is PCB?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Write down the Solving technique of the producer consumer problem with Message passing?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What is system call? Explain the system call process in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
